Lesson 7: Images and simple animation

Pygame CE • S3 onwards • 50-60 min

Learning intention
Load an image and use changing values to create animation.

Success criteria

- I can load and scale an image.
- I can draw it at the player's position.
- I can make something change over time.

Key ideas

- Images are loaded once *before* the loop, never inside it.
- **convert_alpha()** keeps transparency and makes drawing faster.
- Animation means changing something a little every frame - size, position, or which image is shown.

Activities

1. Create an `assets` folder next to the program and add a player image (PNG with transparent background).
2. Replace the player rectangle with the image.
3. Compare the result with the plain shape.
4. Add a simple animation, such as a pulsing target or swapping between two images.

Code

Load and draw an image

```
player_image = pygame.image.load("assets/player.png").convert_alpha()
player_image = pygame.transform.smoothscale(player_image, (SIZE, SIZE))

# in the drawing section:
screen.blit(player_image, (int(x), int(y)))
```

Idea: a pulsing target

```
import math

pulse = 6 * math.sin(pygame.time.get_ticks() / 150)
size = int(SIZE + pulse)
rect = pygame.Rect(0, 0, size, size)
rect.center = target.center
pygame.draw.rect(screen, (240, 190, 60), rect)
```

Idea: two-frame animation

```
frames = [pygame.image.load("assets/player1.png").convert_alpha(),
          pygame.image.load("assets/player2.png").convert_alpha()]

frame = (pygame.time.get_ticks() // 200) % len(frames)
screen.blit(frames[frame], (int(x), int(y)))
```

Checkpoint

The image appears at the right size and follows the player. Collisions still work.

Support and stretch

- **Support:** Supply the image files and the folder layout. Tip: if loading fails, check the program is run from the folder that contains `assets`.
- **Stretch:** Animate between several frames, or flip the image to face the direction of travel.

Reflect

Which is clearer to a player - the shape or the image? Why?

Save your work
Save this version as `lesson7.py` before you start the next lesson.